Mini-Project 4

1 Model Predictive Control of the Inverted Pendulum

Using your model from Mini-Project 3, design a model predictive controller (MPC) for the inverted pendulum about the upright equilibrium, with a limit on the control force. Build the controller in Simulink; MATLAB Function blocks are fine where you need them, but a pure MATLAB script solution is not what we are looking for. Test the controller from the same perturbed upright position used in Mini-Project 3 and compare its response with your LQR controller. In your report, show the resulting pendulum angle and control force and briefly discuss your choice of prediction horizon and force limit.